

Design of polymers with an intrinsic disordered framework for Li-ion conducting solid polymer electrolytes



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ABSTRACT

Polymers with phosphorus in the main chain with intrinsic disordered framework for lithium-ion conductivity were synthesized. Conductivity of one of the solid polymer electrolytes prepared from those polymers with 40 wt% of LiTFSI was $8.9 \times 10^{-4} \text{ S cm}^{-1}$ at 30°C and $5.3 \times 10^{-3} \text{ S cm}^{-1}$ at 80°C . The impressive conductivity is explained by the availability of high conduction pathways provided by frozen chain disorder.

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1. Introduction

Advanced materials and molecules for energy harvesting and storage devices are in demand to improve their performance and efficiency. Case in point, the Li-ion batteries are fast developing as replacement to lead-acid batteries to run automobiles [1]. However, some of its disadvantages – such as high cost and more weight of the Li-ion batteries – and the imperatives to increase its energy density and working temperature necessitate further evolution of new electrode materials and polymer electrolyte systems [2]. Further, there is a need for a thorough understanding of the conduction mechanism to focus on improving ionic conductivities of solid polymer electrolytes [3].

According to the conventional theory, amorphous polymers exhibit ionic conductivity whereas crystalline polymers most likely behave as insulator. The ionic conductivity in amorphous polymer is due to the movement of ions, ensued by the local segmental motion of the polymer chain particularly above its glass transition temperature (T_g). Above T_g , the superstructure of polymers has disordered and flexible environment. Since the polymer chains are in constant motion above its T_g , a vacant space would continually create and disappear, through which an ion move by coordination. It means an ion has to wait to find a new coordination site and a free volume for hopping. This clearly describes that the rate of ion transport depends on local chain dynamics of polymer [4]. Our interest is creating a

fixed perpetual free space through which the ion can move without delay. It would improve the mobility of ion leading to higher ionic conductivity. For that, we have chosen to synthesize materials with a needed structural framework for better lithium ion transport [5].

Many efforts are directed toward synthesizing polymers for lithium ion transport without sacrificing their properties. One such approach is constructing a perfect polymer framework through a crystalline arrangement of polymer chains by either metal organic framework (MOFs) or covalent organic framework (COFs). MOFs and COFs are well known for applications in gas storage, catalysis and molecular separation [6]; but synthesizing the material by MOF and COF approach for lithium ion transport without sacrificing their desired properties is challenging. Alternatively, an amorphous covalent organic framework can be achieved in hyper-cross linked polymers or intrinsic porous polymers. Herein, we report the synthesis of soluble polymers with intrinsic disordered framework supportive of Li-ion conductivity.

2. Experimental section

2.1. Materials and instrumentation

All manipulations involving air and moisture sensitive compounds were carried out using standard Schlenk techniques under dry nitrogen. All solvents to be used under inert atmosphere were thoroughly deoxygenated using freeze–pump–thaw method before use. They were dried and purified by refluxing over a suitable drying agent followed by distillation under nitrogen. The following drying agents were used: sodium benzophenoneketyl

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(hexane, toluene, tetrahydrofuran), P_2O_5 (triethylamine), $Mg(OMe)_2$ (methanol). The compounds, ferrocenylmethyl bis(hydroxymethyl)phosphine sulfide [7] (**M1**), benzyl bis(hydroxymethyl)phosphine sulfide [8] (**M2**) and 1,4-bis(bromomethyl)-2,3,5,6-tetramethylbenzene [9] (**M3**) were synthesized according to the literature procedures. The compound lithium bis(trifluoromethanesulfonyl)imide, $LiN(SO_2CF_3)_2$ purchased from Acros was used as received without purification.

Structures of all monomers and polymers were confirmed by 1H and ^{31}P NMR spectroscopy. All the NMR spectra were recorded on a Bruker Avance 400 MHz FT NMR spectrometer at room temperature. Chemical shifts were reported in parts per million (δ) relative to tetramethylsilane as reference for 1H and ^{13}C NMR. The ^{31}P (162 MHz) spectra were referred to 85% H_3PO_4 . The Netzsch STA 409 PC model was used for thermogravimetric and differential thermal analysis (TG–DTA) to examine the thermal stability. The decomposition behaviour of polymers was studied from 30 °C to 900 °C under the nitrogen flow with a heating rate of 10 °C/min. The temperature of 5% weight loss was chosen as onset point of decomposition (T_d). A Differential Scanning Calorimeter (DSC) from PerkinElmer (Pyris Diamond DSC 8000) was used to find the glass transition temperature (T_g). Measurements were performed at a heating rate of 10 °C/min from 25 °C to 240 °C in case of polymers and 25 °C–280 °C in case of solid polymer electrolytes (**SPEs**). Mass spectra was obtained using Shimadzu–LCMS–2010 instrument operating at an ionizing voltage of 70 eV using EI technique. Molecular weights of the polymers were determined by using Gel Permeation Chromatography (GPC) of Shimadzu 10AVP model equipped with refractive index (RI) detector. The separation was achieved using a Phenogel mixed bed column (300 × 7.80 mm) operated at 30 °C with a flow rate of 0.5 mL/min using tetrahydrofuran (THF) as the eluent and polystyrene as the standard.

2.2. General procedure for the synthesis of polymers P1–P4

To a solution of $RP(S)(CH_2OH)_2$ (7.7 mmol) and 1,4-bis(bromomethyl)-2,3,5,6-tetramethylbenzene (2.46 g, 7.7 mmol) in *N,N*-dimethylacetamide (DMAc) (30 mL), 60% sodium hydride (0.77 g, 19.3 mmol) was added at 0 °C under nitrogen. The mixture was then allowed to warm to 25 °C within 1 h. After 40 h stirring at required temperature, the reaction mixture was poured into hydrochloric acid (3% v/v, 500 mL) to precipitate the polymer as powder. The product was purified by washing with methanol for 24 h using soxhlet apparatus to yield pure polymer.

Polymer 1 (**P1**): Yield 81%. 1H NMR (400 MHz, $CDCl_3$): δ 4.82–4.61 (m, 4H, OCH_2Ph), 4.30–4.03 (m, 9H, Fc), 3.96–3.77 (m, 4H, PCH_2O), 3.28–2.94 (m, 2H, $FcCH_2$), 2.47–2.07 (bs, 12H, CH_3). ^{31}P { 1H } NMR (162 MHz, $CDCl_3$): δ 45.44.

Polymer 2 (**P2**): Yield 72%. 1H NMR (400 MHz, $CDCl_3$): δ 7.67–7.27 (m, 5H, C_6H_5), 4.67–4.36 (m, 4H, OCH_2Ph), 4.18–3.68 (m, 4H, PCH_2O), 3.56–3.32 (m, 2H, $PhCH_2$), 2.47–1.96 (bs, 12H, CH_3). ^{31}P { 1H } NMR (162 MHz, $CDCl_3$): δ 47.94.

Polymer 3 (**P3**): Yield 78%. 1H NMR (400 MHz, $CDCl_3$): δ 4.83–4.56 (m, 4H, OCH_2Ph), 4.34–4.00 (m, 9H, Fc), 3.97–3.69 (m, 4H, PCH_2O), 3.23–2.86 (m, 2H, $FcCH_2$), 2.46–2.05 (bs, 12H, CH_3). ^{31}P { 1H } NMR (162 MHz, $CDCl_3$): δ 44.98 (m), 41.03 (m).

Polymer 4 (**P4**): Yield 75%. 1H NMR (400 MHz, $CDCl_3$): δ 7.76–7.35 (m, 5H, C_6H_5), 4.92–4.49 (m, 4H, OCH_2Ph), 4.03–3.53 (m, 4H, PCH_2O), 3.41–3.00 (m, 2H, $PhCH_2$), 2.42–1.99 (bs, 12H, CH_3). ^{31}P { 1H } NMR (162 MHz, $CDCl_3$): δ 46.19 (m), 42.84 (m).

2.3. Model reaction

To a solution of **M2** (0.20 g, 1 mmol) and benzyl bromide (0.34 g, 2 mmol) in DMAc (2.4 mL) was added sodium hydride (0.05 g,

2 mmol) at 0 °C under nitrogen. The mixture was then allowed to warm to 25 °C within 1 h. After 3 h stirring, methanol (1 mL) was added and the reaction mixture was poured in to water (25 mL). The reaction mixture was extracted with chloroform (20 mL × 3). The organic layer was washed with water (20 mL × 2). Solvent was evaporated to give the crude product, which was purified by column chromatography on silica gel eluting with hexane/ethyl acetate (9: 1). Yield 79%. 1H NMR (400 MHz, $CDCl_3$): δ 7.42–7.32 (m, 10H, C_6H_5), 7.30–7.26 (m, 5H, C_6H_5), 4.65 (dd, J = 21.4 Hz, J = 11.8 Hz, 4H, OCH_2Ph), 3.95–3.84 (m, 4H, OCH_2P), 3.38 (d, J = 13.8 Hz, 2H, PCH_2Ph). ^{13}C { 1H } NMR (100 MHz, $CDCl_3$): δ 136.7, 130.2, 130.1, 128.9, 128.5, 128.4, 128.3, 128.2, 128.1, 127.3, 75.5 (d, J = 10.3 Hz), 67.3 (d, J = 63.9 Hz), 33.5 (d, J = 44.2 Hz). ^{31}P { 1H } NMR (162 MHz, $CDCl_3$): δ 43.50. EI–MS: m/z 397 (M^+ +1).

2.4. Preparation of solid polymer electrolytes SPE1–SPE4

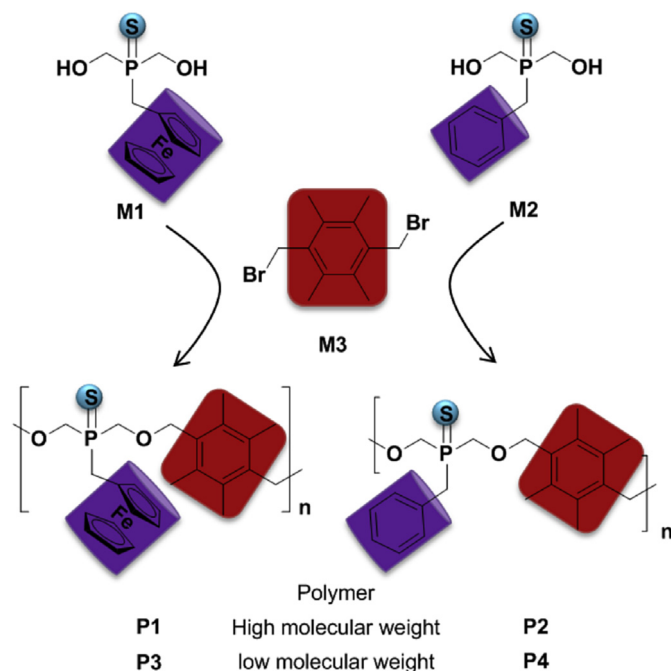
All manipulations were carried out in MBraun glove box filled with ultrapure nitrogen gas. The polymers (**P1–P4**) were dried at 90 °C under vacuum for 8 h. $LiN(SO_2CF_3)_2$ was dried at 150 °C under vacuum for 10 h before making **SPEs**. Initially, the films were made by dissolving the polymers and $LiN(SO_2CF_3)_2$ in a mixture of solvents THF/ $CHCl_3$ at room temperature. However, films of **SPEs** were brittle. Therefore, it was difficult to cut into a proper shape for conductivity studies. Hence, the **SPEs** in the form of pellets were prepared for impedance measurements.

The solid electrolytes with different ratio were prepared as follows. The polymer and lithium salt was ground together inside the glove box and then sealed. Then the flask containing the mixture was tied with a mechanical stirrer rotating at a constant speed (blending) for 24 h. The process was repeated for three times (Total 72 h). The homogeneities of the blended **SPEs** were confirmed as follows: After blending the polymer with lithium salt, samples were taken from the four different regions and their glass transition temperature (T_g) was examined. The T_g of all the samples were matching well with each other. In both cases, the solids were loaded in to a die and then pressed (5 ton) to make pellets of 0.07–0.08 cm thickness and 0.9 cm diameter were obtained for conductivity studies. The **SPE** pellets were sandwiched between two gold plated electrodes housed in a homemade cell.

2.5. Impedance measurements of polymer electrolytes

The impedance measurements and electrochemical stability studies of polymers and **SPEs** were performed in Zahner–Zennium electrochemical workstation with built-in Thales software for data acquisition. The measurements were done in the frequency range of 1 Hz–4 MHz. The measurement cell was constructed in such a way that it can hold specimen (pellet) tightly in order to achieve good interfacial contacts between the electrodes and electrolyte. The measurement cell was sealed completely to avoid contaminations. It was further upheld from the measurements using with different pellets of **SPE1**, which showed the same pattern of results and bulk resistance (R_b).

The conductivities were calculated [10] according to the equation $\sigma = d/(AR_b)$, where d is the thickness of the polymer electrolyte disc, 'A' is the surface area of the pellet and R_b is the bulk resistance value which can be calculated from the intercept of the curve with real axis. Further, an equivalent circuit model was obtained from the computerised least square parameters fitting. Then, the experimental bulk resistance (R_b) was compared with the bulk resistance (R_b) obtained from equivalent circuit model. The consistency of results was checked by repeating three times. The conductivity of blended **SPEs** and **SPEs** derived from evaporation method were in good agreement. For variable temperature



Scheme 1. Synthesis of polymers **P1–P4**. Reaction condition: NaH, DMAc, 60 °C (**P1** and **P2**), 25 °C (**P3** and **P4**).

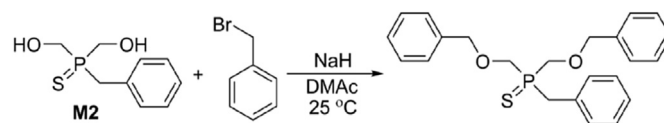
measurements, cell was equilibrated at each temperature for 30 min before the measurements.

The electrochemical stability of SPEs at room temperature was determined using SS|electrolyte|SS cell configuration (SS: gold plated stainless steel electrode) over a wide voltage range (–1 to 8 V) at a scan rate of 5 mV s^{–1}. The samples were sandwiched between gold plated stainless steel electrodes. The third cycle data were used for analysis.

3. Results and discussion

3.1. Synthesis of polyether having phosphorus in main chain

Polyphosphazenes [11a] and polysiloxanes [11b], were good hosts for lithium ion conductivity, accordingly they were reported to have maximum conductivity of 5 × 10^{–5} S cm^{–1} and 4.5 × 10^{–4} S cm^{–1}, respectively. However, their jelly nature hindered the practical use. Polycarbonates [12] were another interesting lithium ion conducting solid polymer host. But they showed a conductivity of 1.5 × 10^{–6} S cm^{–1} which were very low for practicable applications. Alternatively, introduction of phosphorus–carbon bonds, in lieu of carbon–carbon bonds in a polymer chain, could provide disordered arrangement due to an inefficient packing of polymer chains [13,14]. This may lead to interesting physical and chemical properties such as amorphous nature, flame retardant and stability at high temperatures to the polymers [15]. In order to synthesize phosphorus containing polymers; various phosphorus containing monomers are employed. Perhaps, use of



Scheme 2. Synthetic scheme for the model reaction confirming the nucleophilic substitution in the presence of NaH.

substituted bis(hydroxymethyl) phosphine sulfides [16] (Scheme 1) could be an interesting choice to introduce phosphorus–carbon bonds in the main chain of a polymer.

Based on the functional groups present in the monomers (**M1–2** and **M3**), condensation polymerization method was preferred to synthesize the copolymers (Scheme 1). Further, in view of understanding the influence of molecular weight on the ionic conductivities of polymers, we desired to synthesize same polymers of varying molecular weights. The monomers **M1** or **M2** was copolymerized with **M3** (1:1 ratio) in the presence of NaH as a base in dimethylacetamide (DMAc) to yield polymers **P1–P4** (Table 1). The condensation polymers **P1** and **P2** were prepared successfully via nucleophilic substitution reaction at 60 °C whereas the low molecular weight polymers **P3** and **P4** were synthesized at 25 °C. Further increase in reaction temperature led to the formation of insoluble polymers. The crude polymers were purified by washing with methanol in a Soxhlet extractor for 24 h. These reactions yielded novel polyethers wherein the phosphorus atoms are in main chain of the polymers. To confirm the possibility of such nucleophilic substitution reactions, we have carried out a model reaction wherein the monomer **M2** was reacted with benzyl bromide and corresponding product was isolated and characterized (Scheme 2).

All these polymers were solids and readily soluble in chloroform and dimethyl sulfoxide. The ³¹P{¹H} NMR spectra of polymers (**P1**: 45.4 and **P2**: 47.9 ppm) showed a single peak whereas polymers **P3** and **P4** showed two peaks appearing at 41.03, 44.96 ppm (Fig. S10) and at 42.84, 46.18 (Fig. S11) respectively. The difference in ³¹P NMR spectra of high and low molecular weight polymers was due to the chain length. When the length of the polymer chain was small, then the terminal groups were distinguishable from inner groups. Comparing with the chemical shifts of monomers (**M1**: 47.2 ppm and **M2**: 49.2 ppm), the first set of peaks were assigned to the core internal phosphorus atoms of the polymers while the second set of peaks for the terminal group phosphorus atoms of the polymers [17]. The copolymers **P1** and **P2** possess relatively higher molecular weight (*M_w*) than **P3** and **P4** (Table 1).

Thermogravimetric analyses on polymers revealed the thermal stability up to 250 °C (Table 1, Fig. S1). Comparison of **P1** vs. **P3** and **P2** vs. **P4** showed that both *T_d* and *T_g* increased with increasing in chain length (Table 1). The absence of melting temperature (*T_m*) in DSC curves confirmed that the polymers (**P1–P4**) were completely amorphous even with low molecular weight polymers **P3** and **P4** (Fig. S2). The amorphous nature of these polymers further revealed the presence of highly disordered and thereby inefficient packing of polymer chains.

3.2. Solid polymer electrolytes and Li-ion conductivities

Solid polymer electrolytes (SPE1–SPE4) were prepared by blending the polymers (**P1–P4**) with lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) in suitable ratios and lithium ion conductivities were determined. The polymers (**P1–P4**) were dried at 90 °C under vacuum for 8 h. LiN(SO₂CF₃)₂ was dried at 150 °C under vacuum for 10 h before making SPEs. Absence of reaction solvent and moisture in the dried polymer was confirmed from ¹H-NMR

Table 1
Properties of polymers **P1–P4**.

M	T (°C) ^a	P	T _d (°C)	T _g (°C)	M _n (kg mol ^{–1})	M _w (kg mol ^{–1})	PDI
M1	60	P1	288.1	137.0	6.8	11.9	1.74
M2	60	P2	354.4	143.4	9.6	14.6	1.52
M1	25	P3	276.2	130.1	1.8	2.8	1.56
M2	25	P4	250.3	112.3	2.1	4.3	2.03

^a Temperature at which the polymerization was carried out.

Table 2Properties of **SPE1–SPE4** (10%–40%) and their room temperature conductivity.

Conductivity of polymer at room temperature σ (S cm ⁻¹)	P1			P2			P3			P4		
	6.3×10^{-12}			8.1×10^{-12}			3.7×10^{-11}			2.2×10^{-11}		
SPEs (wt% of LiTFSI)	SPE1			SPE2			SPE3			SPE4		
	O:Li	Conductivity σ (S cm ⁻¹)	T_g (°C)	O:Li	Conductivity σ (S cm ⁻¹)	T_g (°C)	Conductivity σ (S cm ⁻¹)	T_g (°C)	Conductivity σ (S cm ⁻¹)	T_g (°C)	Conductivity σ (S cm ⁻¹)	T_g (°C)
SPE (10%)	5.4:1	2.1×10^{-6}	148.7	6.9:1	8.7×10^{-7}	156.4	2.0×10^{-5}	184.3	7.7×10^{-7}	148.9		
SPE (20%)	2.4:1	4.2×10^{-6}	179.3	3.1:1	1.1×10^{-6}	166.5	2.9×10^{-5}	207.8	1.1×10^{-6}	154.5		
SPE (30%)	1.4:1	1.6×10^{-5}	189.1	1.8:1	1.8×10^{-6}	171.0	7.4×10^{-5}	226.4	2.8×10^{-5}	206.6		
SPE (40%)	1.0:1	4.5×10^{-4}	204.8	1.2:1	8.9×10^{-4}	192.8	1.4×10^{-4}	236.1	7.7×10^{-5}	209.6		

spectra of polymers. Electrolytes of varying weight ratio were prepared by the efficient grinding of polymer with lithium salt followed by blending. **SPE1–SPE4** were characterized by using IR, TGA and DSC. The stacked IR spectra of **P1** and **SPE1** with 10–40 wt % of LiTFSI [**SPE1** (10%–40%)] is shown in Fig. S3. IR absorption peak for C–O–C group of **P1** appeared at 1088 cm⁻¹, but the absorption peak for C–O–C group of **SPE1** was shifted to lower frequency (1053 cm⁻¹). The intensity of peak at 1053 cm⁻¹ increased with the increasing wt% of lithium salt. The shift in frequency supported the interaction of C–O–C group with lithium ion [18]. Further, there was no absorption in the spectra matching water confirming the absence of trapped moisture in the **SPEs**.

The TGA analyses (Fig. S4) revealed that thermal stabilities of **SPE1** and **SPE2** remained the same compared with their respective

polymers (**P1** and **P2**); while that of **SPE3** and **SPE4** were increased to 300 °C and 370 °C respectively compared to **P3** and **P4**. The glass transition temperatures (T_g) of **SPEs** (10%–40%) obtained from DSC are shown in Table 2. The absence of melting temperature (T_m) peak of **SPEs** in DSC clearly indicated, after blending with polymers, the LiTFSI did not exist in the form of aggregates (Fig. S5). **SPEs** with 40 wt% of LiTFSI [**SPEs** (40%)] also did not show the peak for T_m , which implied the effective solvation by polymers even with higher wt% of lithium salt. For higher concentration of lithium salts, T_g of **SPEs** was not obvious from DSC graphs even after varying the scanning rates and with more amounts of samples (Fig. S5). The increase in T_g of **SPEs** comparing with respective polymers and difficulty in measuring T_g of **SPEs** with higher salt concentration can be attributed to the effective coordination of oxygen atoms in

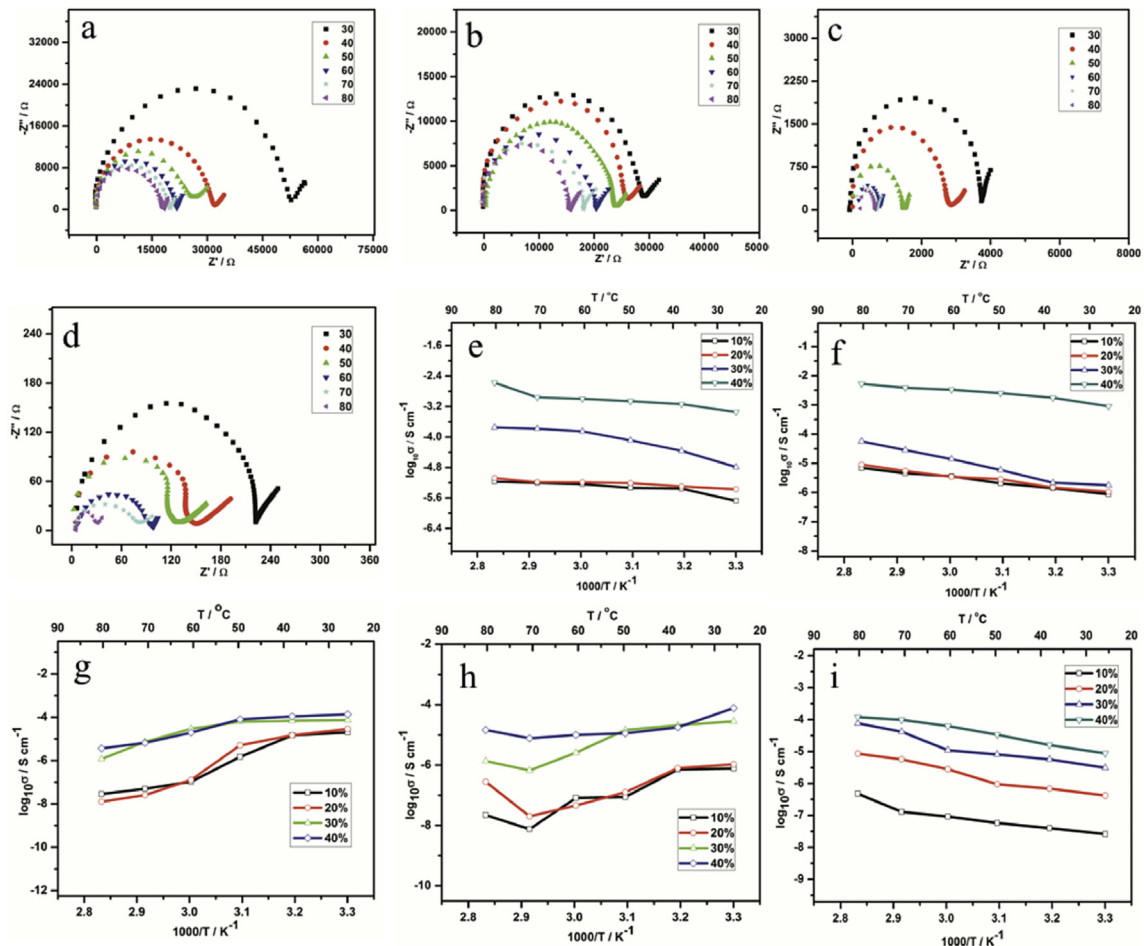


Fig. 1. Temperature dependent Nyquist plots of **SPE1** a) 10% b) 20% c) 30% d) 40%. Arrhenius plots of temperature dependent conductivities of e) **SPE1** f) **SPE2** g) **SPE3** h) **SPE4** i) **SPE4** 2nd heating.

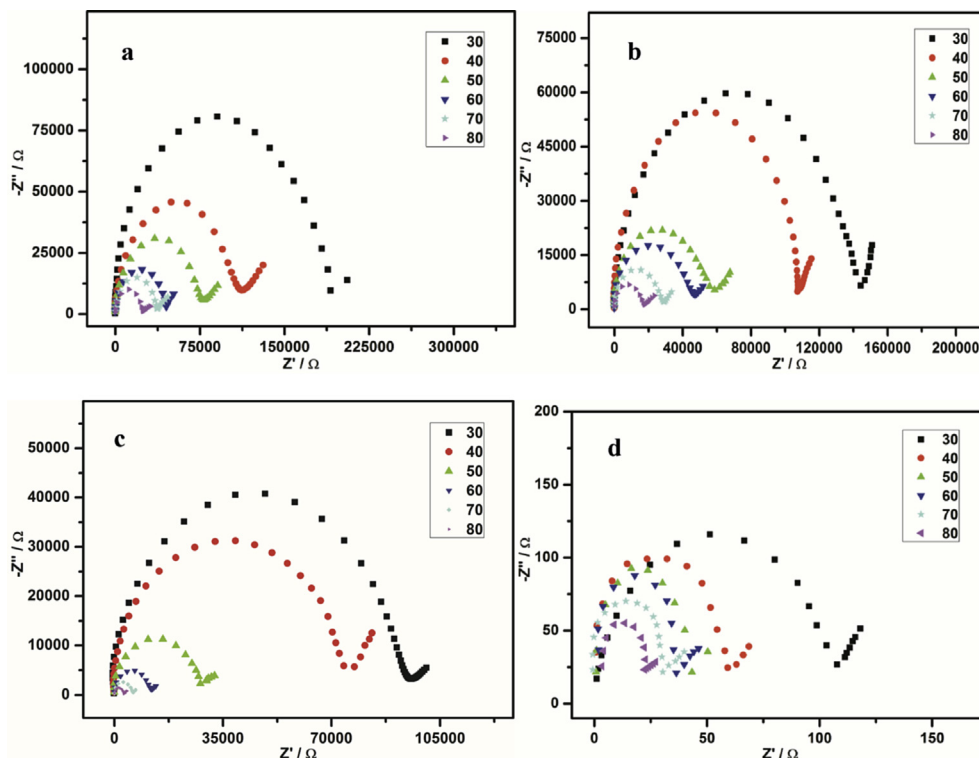


Fig. 2. Temperature dependent Nyquist plots of SPE2 a) 10% b) 20% c) 30% d) 40%.

Table 3
Conductivities of SPE1 (40%)–SPE4 (40%) at increasing temperature.

T (°C)	Conductivity σ (S cm ⁻¹) with 40 wt% of LiTFSI			
	SPE1	SPE2	SPE3	SPE4
30	4.5×10^{-4}	8.9×10^{-4}	1.4×10^{-4}	7.7×10^{-5}
40	7.3×10^{-4}	1.8×10^{-3}	1.1×10^{-4}	1.8×10^{-5}
50	8.7×10^{-4}	2.5×10^{-3}	7.9×10^{-5}	1.1×10^{-5}
60	1.0×10^{-3}	3.3×10^{-3}	2.0×10^{-5}	1.0×10^{-5}
70	1.1×10^{-3}	3.9×10^{-3}	6.6×10^{-6}	7.7×10^{-6}
80	2.7×10^{-3}	5.3×10^{-3}	3.7×10^{-6}	1.5×10^{-5}

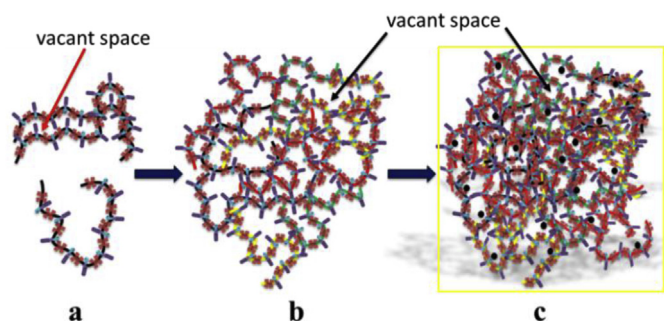


Fig. 3. Schematic representation of a) a fragment of building block polymer chain P1 or P2 b) If the building block grows in 2D c) If the building block grows in 3D (SPE) (Black dots represent lithium-ion).

the polymer chains with lithium ions. With higher salt contents, coordination of polymer with lithium ions could have restricted the stretching and segmental motion of polymer chains. The electrochemical stabilities of SPEs were evaluated (Fig. S7) using SS|electrolyte|SS cell configuration, which showed the anodic limiting potential of SPE1 and SPE2 as 4.8 V and 5.3 V respectively [19].

The conductivities of pure polymers alone were in the region of 10^{-11} S cm⁻¹– 10^{-12} S cm⁻¹ indicating that there was no protonic or electronic conductance. Table 2 depicts the conductivities of SPEs at 30 °C. Among all combinations studied, around room temperature, the SPE2 (40%) exhibited the highest conductivity of 8.9×10^{-4} S cm⁻¹. The Nyquist and Arrhenius plots of temperature dependent conductivities of various SPEs are shown in Figs. 1 and 2. In addition, the conductivities of SPE1 (40%) – SPE4 (40%) at increasing temperature are listed in Table 3. It is evident from Arrhenius plots that when the temperature was increased the conductivities of SPE1 (10%–40%) and SPE2 (10%–40%) were also increasing. The maximum conductivity of 5.3×10^{-3} S cm⁻¹ was attained for SPE2 (40%) at 80 °C. The conductivity of SPE2 (40%) reached the value of 10^{-3} S cm⁻¹ within 40 °C. Conductivities of SPEs were very high compared with their individual polymers. Clear semicircle Nyquist plots suggested that the total conductivity was mainly due to Li⁺ ions only; if some other impurities contributed to the measured conductivity, the plots could have been complicated. The conductivities of SPEs stored in inert conditions for more than six months showed no considerable change. To the best of our knowledge, this is the highest conductivity achieved so far in all solid-state polymer electrolytes reported in literature [13,14,18,20,21]. It is worth mentioning here is that the highest conductivity was achieved in its pure form without any extra additives added to it.

3.3. Reason for high Li-ion conductivity

Generally, amorphous polymers support ion conductivity at a temperature above their T_g. In a normal solid polymer electrolyte (SPE), there were no conductivity observed below T_g. However, our polymers showed good conductivities even at temperatures less than their T_g, nevertheless the properties were much more resembling glassy electrolyte having brittle nature with high T_g.

This observation demanded an alternative explanation [22] to the one based on T_g alone. Angell and coworkers [23] proposed polymer-in-salt electrolytes systems and explained the enhanced conductivity in various Li-polymer electrolytes [24]. Polymer-in-salt systems were characterized by a high salt concentration with much less polymer concentration. This condition lead to a softening of highly associated amorphous salt domains of LiTFSI and similar salts which was the source of enhanced ionic conductivity. However, the polymer concentration was more in our solid electrolytes than the Li-ion concentration. Therefore, the observed high conductivities is explained proposing an inherent disorder in the system as follows.

The polymers **P1–P4** were produced using flexible monomers (**M1** and **M2**) having bulky rigid pendant groups and a steric rigid co-monomer (**M3**). Repetitive combination of flexible monomers and steric rigid co-monomers could yield unsymmetrical shape to polymer chains. Further, the steric control of bulky rigid pendant groups *i.e.* ferrocenylmethyl (**P1** and **P3**) or benzyl (**P2** and **P4**), as well as a rigid tetramethylbenzene unit (**M3**) could restrict the polymer chains coming closer and prevent tight folding of polymer chains within or between them. Additionally, the rigid bulky groups could also restrict the rotational freedom of polymer backbone. All these circumstances collectively resulting disordered arrangement and inefficient packing of polymer chains. Fig. 3 schematically explains the formation of free space by polymer chains. When polymer chains with restricted segmental motion are packing, a space would create (Fig. 3a). As several polymer chains are arranged, it can form perpetual interconnected vacant space (Fig. 3b and c). The interconnected perpetual free volume would introduce intrinsic fixed conduction pathway inside the polymer matrix (Fig. 3c). However, at the highest salt concentrations, the ratio Li:O is near 1:1 and thus a classical solution by formation of lithium complexes with 3–4 donor atoms is uncertain. Therefore, though the intrinsic fixed conduction pathway inside the polymer matrix seems helping the seamless mobility of Li ions, in order to confirm the mechanism of high conductivity, one needs more evidence [13].

To understand the ion conductivity further, we have measured temperature dependent conductivities of all SPEs (Table 3). Normally, local segmental motion of polymer chains increases with increasing temperature leading to increase in conductivity. As expected, conductivities of **SPE1** (10%–40%) and **SPE2** (10%–40%) increased with increasing temperature. Similar results were obtained in second and third cycle heating of **SPE1** and **SPE2**. Though these temperature dependent conductivities of SPEs did not strictly follow either Arrhenius or VTF behaviours [25], the order of magnitude of the measured conductivities at 30 °C is attracting.

In contrast, the conductivity of **SPE3** (10%–40%) and **SPE4** (10%–40%) decreased with increase in temperature (Table 3 and Fig. 1g and h). It suggested a secondary influence on ionic conduction due to a non-stable polymeric network. It can be explained as follows; since the polymer chain lengths of **P3** and **P4** were smaller, it was possible that the polymer chains can stretch and unfold easily with increasing temperature and begin to rearrange in such a way destroying the framework. Since the intrinsic fixed free space were disconnected, which was the responsible for effortless mobility, resulting chaotic conductivities at higher temperatures. Nevertheless, the polymers **P1** and **P2** presumably retain their framework even at higher temperatures because of longer chain lengths. With **SPE4** (40%), the conductivity started increasing after 70 °C and attained the value of $1.5 \times 10^{-5} \text{ S cm}^{-1}$ at 80 °C (Fig. 1h). After the measurements at 80 °C, the **SPE4** (10%–40%) sample was allowed to cool to room temperature and the measurements were repeated from 30 to 80 °C. It was found that the conductivities of **SPE4** (10%–40%) increased with increasing temperature (Fig. 1i) probably because of the mechanism described elsewhere [25].

4. Conclusions

In conclusion, other than poly(ethylene oxide), only few polymers like polycarbonates [12] were shown as a potential candidate for lithium-ion polymer electrolytes; however, their ionic conductivities were too low at room temperature. The synthesized phosphorus containing polymers **P1–P4** were highly stable polymers, solids at room temperature and completely amorphous in nature. The **SPE2** (40%) showed the highest conductivity of $8.9 \times 10^{-4} \text{ S cm}^{-1}$ at 30 °C, which demonstrated the significance of structural arrangement of polymer framework. The impressive conductivities may be explained by the intrinsic fixed free space could be generated by the inefficient packing of polymer chains while assemblage. The working voltage range of SPEs was in the range from 0 V to 5.3 V, which was substantially high for the Li-ion batteries concern. Apart from these, presence of phosphorus in these polymers would bring flame-retardant property to SPEs [26].

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.polymer.2015.08.004>.

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